# Area and volume — Part 1 answer key

| # | ID | Answer | Difficulty |
|---|---|---|---|
| 1.1 | c53570c0-9d6b-49d6-8b3f-c45a4ef532db | B | Easy |
| 1.2 | 7fb7e2c6-237c-42b9-a178-4810e8c6cd93 | D | Easy |
| 1.3 | 5d25f220-0193-4939-a687-673246d3076b | B | Easy |
| 1.4 | cdf507c8-4e6e-437e-bae9-d9f6a9e92f13 | 880 | Easy |
| 1.5 | cde76af2-5837-4a16-aae2-109dd079e384 | D | Easy |
| 1.6 | a3a441b7-c6ac-48e3-9e5c-759686bbc5ee | C | Easy |
| 1.7 | 2bde9f21-d49c-42af-b7ec-38c87b242d02 | 3 | Easy |
| 1.8 | 5a173e5b-ac90-4f05-806e-fe38d80fcfd9 | D | Easy |
| 1.9 | 2fcb111d-48d0-4ee3-925e-0ffeec3615a4 | A | Easy |
| 1.10 | 9a1d9f4a-bbb7-4e9d-bb0c-c5445263d935 | B | Easy |
| 1.11 | 4a39e52e-02ea-49d7-a695-87a667bdb785 | A | Easy |
| 1.12 | 05f34b6c-f496-4574-bc7b-cffaaa469975 | C | Easy |
| 1.13 | 21882da9-ffd9-4340-8877-b77ba9972851 | C | Easy |
| 1.14 | 7ce3a2e8-bebb-4040-84e7-3f30170327e3 | 2048 | Easy |
| 1.15 | c2922131-fa53-47c8-b19a-4f809b49ffe3 | 986 | Easy |
| 1.16 | 77d41ee0-634a-4578-a6b0-6ec8687d2c0a | D | Easy |
| 1.17 | 025774-DC | B | Easy |
| 1.18 | f5e0aefd-1908-42ec-9072-52390bef3063 | D | Easy |
| 1.19 | 76f07394-0a95-4e65-86b8-e42618d65c92 | B | Easy |
| 1.20 | 21b61a47-8395-41b7-9031-ab756ca63c22 | C | Easy |
| 1.21 | 1e523d74-12a9-4f3a-8279-13ec77276c13 | 180 | Easy |
| 1.22 | 8ec98031-aeed-4459-a154-8c48f0ec122a | B | Easy |

| | | | |
|---|---|---|---|
| 1.23 | 786b1f64-6764-44c0-a7d6-75880720fc77 | A | Easy |
| 1.24 | ca7c086a-76e8-4959-b23f-94d448a1c1c3 | D | Easy |
| 1.25 | f06ed67a-28a2-4bfe-9834-47786c3571e8 | B | Easy |
| 1.26 | b2842dd1-e205-4ff0-a977-6379b5655857 | A | Easy |
| 1.27 | 09ef21b2-86ef-4977-8eef-37b47c81624e | D | Easy |
| 1.28 | 8cb6a96f-5086-4d10-8015-3085accf0ee2 | D | Easy |
| 1.29 | 60500dd8-893a-4fea-903d-de896120774c | C | Easy |
| 1.30 | cab19268-2903-44cd-aa02-6f253b7a4efc | A | Easy |
| 1.31 | 0c4e91ab-c388-4002-8b80-16bf6f098d01 | A | Easy |
| 1.32 | 027084-DC | A | Easy |
| 1.33 | a5cc3d06-5c10-40d6-bfec-b01230ec82b8 | C | Easy |
| 1.34 | f9aa6eb4-a501-4825-9c17-1efda9ecd0cf | C | Easy |
| 1.35 | 7b4cffc9-9565-413b-810d-67d7f92aa56e | 1800 | Easy |
| 1.36 | 2cc3f845-5222-4c22-8764-9d19c7b188fb | A | Easy |
| 1.37 | 13e26f81-104e-4cc5-b4b5-52d4234d4b53 | 36 | Medium |
| 1.38 | bf7a641a-e835-4a95-a957-157c3c91e4ca | D | Medium |
| 1.39 | 9f8442fc-4e43-48a4-9eda-fd446fd06043 | 29 | Medium |
| 1.40 | 2576e4c1-c4ee-4f32-8350-159f08967b70 | A | Medium |
| 1.41 | f59b253c-f13a-4027-850d-9d1abc35bde5 | 4205 | Medium |
| 1.42 | 029396-DC | C | Medium |
| 1.43 | 7ffcadaa-3611-4602-92ad-c14085a27aef | 4.41, 441/100 | Medium |
| 1.44 | 357b0938-dcf1-4984-934a-7d55d036717a | 27 | Medium |
| 1.45 | 7c5d80a9-e593-4cf8-b36b-bd89ed6226c7 | B | Medium |
| 1.46 | 86cc8eb8-5cbf-4188-9abd-1839bfec0526 | 126 | Medium |
| 1.47 | 070925-DC | D | Medium |
| 1.48 | 924fda1b-aa0e-4da6-bd05-7a55e96639dd | B | Medium |

| | | | |
|---|---|---|---|
| 1.49 | 66a68c59-a01d-4c33-b54e-5a44e513c248 | D | Medium |
| 1.50 | 06614-DC | B | Medium |
| 1.51 | a6f2f98a-1452-4d23-aa35-d397a85c694f | A | Medium |
| 1.52 | 22accec5-7fd4-4bbf-8af1-a6077fe56fc3 | A | Medium |
| 1.53 | ea4fb9f5-47ef-4d74-89e2-5f8e44710904 | C | Medium |
| 1.54 | 39d02567-fb2f-4a7c-a802-69186c6a894f | 31 | Medium |
| 1.55 | 00113-DC | C | Medium |

## Rationales

▼ 1.1 — B

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side *AB* and side *AC* are $4$ inches and $7$ inches, respectively. Let $x$ represent the length, in inches, of side *BC*. The sum of the lengths, in inches, of all three sides of triangle *ABC* can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle *ABC* is $17$ inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting $11$ from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side *BC* is $6$.

Choice A is incorrect. This is the length, in inches, of side *AB*.

Choice C is incorrect. This is the length, in inches, of side *AC*.

Choice D is incorrect. This is the sum of the lengths, in inches, of sides *AB* and *AC*.

▼ 1.2 — D

Choice D is correct. The area of a rectangle is given by $lw$, where $l$ is the length of the rectangle and $w$ is the width of the rectangle. It's given that a rectangle has a length of $3$ units and a width of $39$ units. It follows that the area of the rectangle is $3 \cdot 39$ square units. Therefore, the expression that gives the area, in square units, of this rectangle, is $3 \cdot 39$.

Choice A is incorrect. This expression gives the perimeter, in units, of this rectangle.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.3 — B

Choice B is correct. The area $A$ of a rectangle is the product of its length $l$ and its width $w$. It follows that $A = lw$. It's given that rectangle $ABCD$ has an area of $280$ square centimeters and the length of the rectangle is $10$ centimeters. Substituting $280$ for $A$ and $10$ for $l$ in the equation $A = lw$ yields $280 = 10w$. Dividing both sides of this equation by $10$ yields $28 = w$. Therefore, the width, in centimeters, of the rectangle is $28$.

Choice A is incorrect. This is the length, in centimeters, not the width, in centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is half the area, in square centimeters, not the width, in centimeters, of the rectangle.

▼ 1.4 — 880

The correct answer is $880$. The volume, $V$, of a right rectangular prism is given by the formula $V = lwh$, where $l$ is the length, $w$ is the width, and $h$ is the height of the prism. It's given that a right rectangular prism has a length of $11$ meters, a width of $8$ meters, and a height of $10$ meters. Substituting $11$ for $l$, $8$ for $w$, and $10$ for $h$ in the formula $V = lwh$ yields $V = (11)(8)(10)$, or $V = 880$. Therefore, the volume, in cubic meters, of the prism is $880$.

▼ 1.5 — D

Choice D is correct. The perimeter of a figure is equal to the sum of the measurements of the sides of the figure. It's given that the rectangle has a length of $4$ inches and a width of $9$ inches. Since a rectangle has $4$ sides, of which opposite sides are parallel and equal, it follows that the rectangle has two sides with a length of $4$ inches and two sides with a width of $9$ inches. Therefore, the perimeter of this rectangle is $4 + 4 + 9 + 9$, or $26$ inches.

Choice A is incorrect. This is the sum, in inches, of the length and the width of the rectangle.

Choice B is incorrect. This is the sum, in inches, of the two lengths and the width of the rectangle.

Choice C is incorrect. This is the sum, in inches, of the length and the two widths of the rectangle.

▼ 1.6 — C

Choice C is correct. The perimeter of a triangle is the sum of the lengths of the three sides of the triangle. It's given that the length of one side of an equilateral triangle is $28$ inches. Since the lengths of the three sides of an equilateral triangle are equal, it follows that the perimeter, in inches, of this equilateral triangle is $28 + 28 + 28$, or $84$.

Choice A is incorrect. This is the length of one side, not the perimeter, of this equilateral triangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the perimeter of a square, not an equilateral triangle, with side lengths of $28$ inches.

▼ 1.7 — 3

The correct answer is $3$. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It's given that the area of the rectangle is $57$ square inches and the length of the longest side of the rectangle is $19$ inches. Let $x$ represent the length, in inches, of the shortest side of this rectangle. It follows that $57 = 19x$. Dividing both sides of this equation by $19$ yields $3 = x$. Therefore, the length, in inches, of the shortest side of the rectangle is $3$.

▼ 1.8 — D

Choice D is correct. The area, $A$, of a rectangle can be found using the formula $A = lw$, where $l$ represents the length and $w$ represents the width of the rectangle. It's given that a rectangle has a length of $56$ inches and a width of $28$ inches. Substituting $56$ for $l$ and $28$ for $w$ in the formula $A = lw$ yields $A = (56)(28)$, or $A = 1{,}568$. Therefore, the area, in square inches, of the rectangle is $1{,}568$.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.9 — A

Choice A is correct. The perimeter of a triangle is the sum of the lengths of its three sides. The triangle shown has side lengths $x$, $y$, and $z$. It's given that the triangle has a perimeter of $22$ units. Therefore, $x + y + z = 22$. If $x = 9$ units and $y = 7$ units, the value of $z$, in units, can be found by substituting $9$ for $x$ and $7$ for $y$ in the equation $x + y + z = 22$, which yields $9 + 7 + z = 22$, or $16 + z = 22$. Subtracting $16$ from both sides of this equation yields $z = 6$. Therefore, if $x = 9$ units and $y = 7$ units, the value of $z$, in units, is $6$.

Choice B is incorrect. This is the value of $y$, in units, not the value of $z$, in units.

Choice C is incorrect. This is the value of $x$, in units, not the value of $z$, in units.

Choice D is incorrect. This is the value of $x + y$, in units, not the value of $z$, in units.

▼ 1.10 — B

Choice B is correct. The formula for the area, $A$, of a triangle is $A = \frac{1}{2}bh$, where $b$ is the length of the base of the triangle and $h$ is the height of the triangle. It's given that the length of the base of a triangle is $8$ inches and the height of the triangle is $12$ inches. Substituting $8$ for $b$ and $12$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}812$, or $A = 48$. Therefore, the area, in square inches, of the triangle is $48$.

Choice A is incorrect. This is the sum of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the product of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

▼ 1.11 — A

Choice A is correct. It's given that the area of a square is $64$ square inches. The area $A$, in square inches, of a square is given by the formula $A = s^2$, where $s$ is the side length, in inches, of the square. Substituting $64$ for $A$ in this formula yields $64 = s^2$. Taking the positive square root of both sides of this equation yields $8 = s$. Thus, the side length, in inches, of this square is $8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.12 — C

Choice C is correct. The area, $A$, of a rectangle is given by the formula $A = lw$, where $l$ represents the length of the rectangle and $w$ represents its width. It's given that the rectangle has a length of $7$ inches and a width of $6$ inches. Substituting $7$ for $l$ and $6$ for $w$ in the formula $A = lw$ yields $A = 76$, or $A = 42$. Thus, the area, in square inches, of the rectangle is $42$.

Choice A is incorrect. This is the sum, not the product, of the length and width of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is twice the area, in square inches, of the rectangle.

▼ 1.13 — C

Choice C is correct. The area, $A$, of a rectangle is given by the formula $A = lw$, where $l$ is the length and $w$ is the width of the rectangle. It's given that rectangle $ABCD$ shown has a length of $7$ units and a width of $2$ units. Substituting $7$ for $l$ and $2$ for $w$ in the formula $A = lw$ yields $A = (7)(2)$, or $A = 14$. Therefore, the area, in square units, of rectangle $ABCD$ is $14$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.14 — 2048

The correct answer is $2{,}048$. The area $A$, in square inches, of a rectangle is equal to the product of its length $l$, in inches, and its width $w$, in inches, or $A = lw$. It's given that the rectangle has a length of $64$ inches and a width of $32$ inches. Substituting $64$ for $l$ and $32$ for $w$ in the equation $A = lw$ yields $A = 6432$, or $A = 2{,}048$. Therefore, the area, in square inches, of the rectangle is $2{,}048$.

▼ 1.15 — 986

The correct answer is $986$. The area, $A$, of a rectangle is given by $A = lw$, where $l$ is the length of the rectangle and $w$ is its width. It's given that the length of the rectangle is $34$

centimeters (cm) and the width is 29 cm. Substituting 34 for $l$ and 29 for $w$ in the equation $A = lw$ yields $A = 3429$, or $A = 986$. Therefore, the area, in square centimeters, of this rectangle is 986.

▼ 1.16 — D

Choice D is correct. It's given that rectangle P has an area of 72 square inches. If a rectangle with an area of 20 square inches is removed from rectangle P, the area, in square inches, of the resulting figure is $72 - 20$, or 52.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.17 — B

Choice B is correct. Since triangles ABC and DEF are similar and $DE = 2AB$, the length of each side of triangle DEF is two times the length of its corresponding side in triangle ABC. Therefore, the perimeter of triangle DEF is two times the perimeter of triangle ABC. Since the perimeter of triangle ABC is 20, the perimeter of triangle DEF is 40.

Choice A is incorrect. This is half, not two times, the perimeter of triangle ABC. Choice C is incorrect. This is two times the perimeter of triangle DEF rather than two times the perimeter of triangle ABC. Choice D is incorrect. This is six times, not two times, the perimeter of triangle ABC.

▼ 1.18 — D

Choice D is correct. The area $A$, in square centimeters, of a rectangle can be found using the formula $A = lw$, where $l$ is the length, in centimeters, of the rectangle and $w$ is its width, in centimeters. It's given that the rectangle has a length of 36 centimeters and a width of 34 centimeters. Substituting 36 for $l$ and 34 for $w$ in the formula $A = lw$ yields $A = 3634$, or $A = 1{,}224$. Therefore, the area, in square centimeters, of this rectangle is 1,224.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, in centimeters, not the area, in square centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.19 — B

Choice B is correct. The area of a rectangle with length $l$ and width $w$ can be found using the formula $A = lw$. It's given that the rectangle has a length of $4 \text{ cm}$ and a width of $2 \text{ cm}$. Therefore, the area of this rectangle is $4 \text{ cm}(2 \text{ cm})$, or $8 \text{ cm}^2$.

Choice A is incorrect. This is the sum, $\text{in cm}$, of the length and width of the rectangle, not the area, $\text{in cm}^2$.

Choice C is incorrect. This is the perimeter, $\text{in cm}$, of the rectangle, not the area, $\text{in cm}^2$.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

▼ 1.20 — C

Choice C is correct. The area $A$, $\text{in square centimeters } (\text{cm}^2)$, of a square with side length $s$, $\text{in cm}$, is given by the formula $A = s^2$. It's given that the square has a side length of $55 \text{ cm}$. Substituting $55$ for $s$ in the formula $A = s^2$ yields $A = 55^2$, or $A = 3{,}025$. Therefore, the area, $\text{in cm}^2$, of the square is $3{,}025$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, $\text{in cm}$, of the square, not its area, $\text{in cm}^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.21 — 180

The correct answer is $180$. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of $45$. Since a square is a polygon with $4$ sides, the perimeter of this square is $45 + 45 + 45 + 45$, or $180$.

▼ 1.22 — B

Choice B is correct. The area of a square is $s^2$, where $s$ is the side length of the square. It's given that the length of a side of square X is $9$ centimeters. It follows that the area of square X is $9^2$, or $81$ square centimeters. It's also given that the area of rectangle Y is $32$ square centimeters. Therefore, the total area, in square centimeters, of square X and rectangle Y is $81 + 32$, or $113$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area of square X, not the total area of square X and rectangle Y, in square centimeters.

▼ 1.23 — A

Choice A is correct. The volume, $V$, of a rectangular prism is given by the formula $V = lwh$, where $l$ is the length of the base, $w$ is the width of the base, and $h$ is the height of the prism. It's given that each base of a right rectangular prism has a length of $19$ inches and a width of $8$ inches, and that the prism has a volume of $2{,}736$ cubic inches. Substituting $19$ for $l$, $8$ for $w$, and $2{,}736$ for $V$ in the formula $V = lwh$ gives $2{,}736 = (19)(8)(h)$, or $2{,}736 = 152h$. Dividing each side of this equation by $152$ yields $18 = h$. Therefore, the height, in inches, of the prism is $18$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

▼ 1.24 — D

Choice D is correct. It's given that the width of this rectangle is $7$ centimeters and that the length of this rectangle is $40$ centimeters longer than the width. Therefore, the length of this rectangle is $7 + 40$, or $47$, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is $747$, or $329$.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.25 — B

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three of its sides. It's given that the lengths of two sides of a triangle are $4$ centimeters and $6$ centimeters. Let $x$ represent the length, in centimeters, of the third side of this triangle. The sum of the lengths, in centimeters, of all three sides of the triangle can be represented by the expression $4 + 6 + x$. Since it's given that the perimeter of the triangle is $18$ centimeters, it follows that $4 + 6 + x = 18$, or $10 + x = 18$. Subtracting $10$ from both sides of this equation yields $x = 8$. Therefore, the length, in centimeters, of the third side of this triangle is $8$.

Choice A is incorrect. If the length of the third side of this triangle were $2$ centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 2$, or $12$, not $18$.

Choice C is incorrect. If the length of the third side of this triangle were $10$ centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 10$, or $20$, not $18$.

Choice D is incorrect. If the length of the third side of this triangle were $24$ centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 24$, or $34$, not $18$.

▼ 1.26 — A

Choice A is correct. It's given that triangle *JKL* is similar to triangle *RST*, where *J* corresponds to *R* and *K* corresponds to *S*. It follows that $\bar{JK}$ corresponds to $\bar{RS}$. If two triangles are similar, then the scale factor between their perimeters is equal to the scale factor between the lengths of their corresponding sides. It's given that the length of $\bar{JK}$ is $15$ and the length of $\bar{RS}$ is $135$. Therefore, the scale factor from the length of $\bar{JK}$ to the length of $\bar{RS}$ is $\frac{135}{15}$, or $9$. It's given that the perimeter of triangle *JKL* is $36$. Let $p$ represent the perimeter of triangle *RST*. It follows that $\frac{p}{36} = 9$. Multiplying each side of this equation by $36$ yields $p = 324$. Therefore, the perimeter of triangle *RST* is $324$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.27 — D

Choice D is correct. It's given that triangle R has an area of $80\ \text{cm}^2$. The area of a square is $l^2$, where $l$ is the side length of the square. It's given that square S has side lengths of $4\ \text{cm}$. It follows that the area, in $\text{cm}^2$, of square S is $4^2$, or $16$. Therefore, the total area, in $\text{cm}^2$, of triangle R and square S is $80 + 16$, or $96$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.28 — D

Choice D is correct. The area of a square is given by $s^2$, where $s$ is the length of each side of the square. It's given that the length of each side of a square is $35$ inches. Substituting $35$ for $s$ in the expression $s^2$ yields $35^2$. Therefore, the expression that represents the area, in square inches, of the square is $35^2$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression represents the perimeter, in inches, of the square rather than the area, in square inches.

▼ 1.29 — C

Choice C is correct. The perimeter of a quadrilateral is the sum of the lengths of its four sides. It's given that the rectangle has a length of $13$ and a width of $6$. It follows that the rectangle has two sides with length $13$ and two sides with length $6$. Therefore, the perimeter of the rectangle is $13 + 13 + 6 + 6$, or $38$.

Choice A is incorrect. This is the sum of the lengths of the two sides with length $6$, not the sum of the lengths of all four sides of the rectangle.

Choice B is incorrect. This is the sum of the lengths of the two sides with length $13$, not the sum of the lengths of all four sides of the rectangle.

Choice D is incorrect. This is the perimeter of a rectangle that has four sides with length $13$, not two sides with length $13$ and two sides with length $6$.

▼ 1.30 — A

Choice A is correct. The area, $A$, of a triangle is given by the formula $A = \frac{1}{2}bh$, where $b$ is the length of the base of the triangle and $h$ is the height of the triangle. In the right triangle shown, the length of the base of the triangle is $9$ units, and its height is $4$ units. Substituting

$9$ for $b$ and $4$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(9)(4)$, or $A = 18$. Therefore, the area of the triangle, in square units, is $18$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of the base of the triangle, in units, not the area of the triangle, in square units.

Choice D is incorrect. This is the height of the triangle, in units, not the area of the triangle, in square units.

▼ 1.31 — A

Choice A is correct. The area $A$, in square meters, of a rectangle is the product of its length $l$, in meters, and its width $w$, in meters; thus, $A = lw$. It's given that a rectangle has an area of $63$ square meters and a length of $9$ meters. Substituting $63$ for $A$ and $9$ for $l$ in the equation $A = lw$ yields $63 = 9w$. Dividing both sides of this equation by $9$ yields $7 = w$. Therefore, the width, in meters, of the rectangle is $7$.

Choice B is incorrect. This is the difference between the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Choice C is incorrect. This is the square of the length, in meters, not the width, in meters, of the rectangle.

Choice D is incorrect. This is the product of the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

▼ 1.32 — A

Choice A is correct. It's given that the volume of a right rectangular prism is $\ell wh$. The prism shown has a length of 6 cm, a width of 2 cm, and a height of 3 cm. Thus, $\ell wh = (6)(2)(3)$, or 36 cubic centimeters.

Choice B is incorrect. This is the volume of a rectangular prism with edge lengths of 6, 2, and 2. Choice C is incorrect and may result from only finding the product of the length and width of the base of the prism. Choice D is incorrect and may result from finding the sum, not the product, of the edge lengths of the prism.

▼ 1.33 — C

Choice C is correct. The area of a rectangle with length $l$ and width $w$ can be found using the formula $A = lw$. It's given that the rectangle has a length of $17 \text{ cm}$ and a width of $7 \text{ cm}$. Therefore, the area of this rectangle is $A = 177$, or $119 \text{ cm}^2$.

Choice A is incorrect. This is the sum of the length and width of the rectangle, not the area.

Choice B is incorrect. This is the perimeter of the rectangle, not the area.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

▼ 1.34 — C

Choice C is correct. It's given that $x = 7$ centimeters and $y = 5$ centimeters for the rectangle. The area, $A$, of a rectangle can be found using the formula $A = lw$, where $l$ represents the length and $w$ represents the width. In this rectangle, $x$ represents the length and $y$ represents the width. Substituting $7$ for $l$ and $5$ for $w$ in the equation yields $A = 75$, or $A = 35$. Therefore, the area of the rectangle is $35$ square centimeters.

Choice A is incorrect. This is the length, not the area, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.35 — 1800

The correct answer is $1{,}800$. The area, $A$, of a triangle can be found using the formula $A = \frac{1}{2}bh$, where $b$ is the base length of the triangle and $h$ is the height of the triangle. It's given that the triangle has a base length of $40$ centimeters and a height of $90$ centimeters. Substituting $40$ for $b$ and $90$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}4090$, or $A = 1{,}800$. Therefore, the area, in square centimeters, of the triangle is $1{,}800$.

▼ 1.36 — A

Choice A is correct. The area $A$, in square inches, of a rectangle is the product of its length $l$, in inches, and its width $w$, in inches; thus, $A = lw$. It's given that the area of a rectangle is $630$ square inches and the length of the rectangle is $70$ inches. Substituting $630$ for $A$ and $70$ for $l$ in the equation $A = lw$ yields $630 = 70w$. Dividing both sides of this equation by $70$ yields $9 = w$. Therefore, the width, in inches, of this rectangle is $9$.

Choice B is incorrect. This is the length, not the width, in inches, of the rectangle.

Choice C is incorrect. This is half the area, in square inches, not the width, in inches, of the rectangle.

Choice D is incorrect. This is the difference between the area, in square inches, and the length, in inches, of the rectangle, not the width, in inches, of the rectangle.

▼ 1.37 — 36

The correct answer is $36$. The area, $A$, of a triangle is given by the formula $A = \frac{1}{2}bh$, where $b$ represents the length of the base of the triangle and $h$ represents its height. It's given that the triangle has an area of $36$ square units and a height of $2$ units. Substituting $36$ for $A$ and $2$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $36 = \frac{1}{2}b(2)$, or $36 = b$. Thus, the length, in units, of the base of this triangle is $36$.

▼ 1.38 — D

Choice D is correct. The perimeter, $P$, of a square can be found using the formula $P = 4s$, where $s$ is the length of each side of the square. It's given that square X has a side length of $12$ centimeters. Substituting $12$ for $s$ in the formula for the perimeter of a square yields $P = 412$, or $P = 48$. Therefore, the perimeter of square X is $48$ centimeters. It's also given that the perimeter of square Y is $2$ times the perimeter of square X. Therefore, the perimeter of square Y is $248$, or $96$, centimeters. Substituting $96$ for $P$ in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by $4$ gives $24 = s$. Therefore, the length of one side of square Y is $24$ centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.39 — 29

The correct answer is $29$. The volume, $V$, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where $r$ is the radius of the base of the cylinder and $h$ is the height of the cylinder. Since the base of the cylinder is a circle with radius $r$, the area of the base of the cylinder is $\pi r^2$. It's given that a right circular cylinder has a volume of $377$ cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is $13$ square centimeters; therefore, $\pi r^2 = 13$. Substituting $377$ for $V$ and $13$ for $\pi r^2$ in the formula $V = \pi r^2 h$ yields

$377 = 13h$. Dividing both sides of this equation by $13$ yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is $29$.

▼ 1.40 — A

Choice A is correct. The area of a square is $s^2$, where $s$ is the side length of the square. Therefore, the area of square B is $b^2$, where $b$ is the side length of square B. It's given that square A has side lengths that are $246$ times the side lengths of square B. Therefore, the side length of square A can be represented by the expression $246b$. It follows that the area of square A is $(246b)^2$, or $60{,}516b^2$. It's given that the area of square A is $k$ times the area of square B, so $60{,}516b^2 = kb^2$. Therefore, the value of $k$ is $60{,}516$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.41 — 4205

The correct answer is $4{,}205$. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of $5$ congruent square faces. It's also given that the length of each edge is $29$ inches. Let $s$ represent the length of an edge of a square. It follows that the area of a square is equal to $s^2$. Therefore, the area of each of the $5$ square faces is equal to $29^2$, or $841$, square inches. Since the box consists of $5$ congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is $5841$, or $4{,}205$, square inches.

▼ 1.42 — C

Choice C is correct. The area A of a circle with radius r is given by the formula $A = \pi r^2$. Thus, a circle with radius 2 has area $\pi(2^2)$, which can be rewritten as $4\pi$. The area of a square with side length s is given by the formula $A = s^2$. Thus, if a square has the same area as a circle with radius 2, then $s^2 = 4\pi$. Since the side length of a square must be a positive number, taking the square root of both sides of $s^2 = 4\pi$ gives $s = \sqrt{4\pi}$. Using the properties of square roots, $\sqrt{4\pi}$ can be rewritten as $(\sqrt{4})(\sqrt{\pi})$, which is equivalent to $2\sqrt{\pi}$. Therefore, $s = 2\sqrt{\pi}$.

Choice A is incorrect. The side length of the square isn't equal to the radius of the circle. Choices B and D are incorrect and may result from incorrectly simplifying the expression $\sqrt{4\pi}$.

▼ 1.43 — 4.41, 441/100

The correct answer is $4.41$. The area, $A$, of a circle is given by the formula $A = \pi r^2$, where $r$ is the radius of the circle. It's given that the area of the circle is $b\pi$ square inches, where $b$ is a constant, and the radius of the circle is $2.1$ inches. Substituting $b\pi$ for $A$ and $2.1$ for $r$ in the formula $A = \pi r^2$ yields $b\pi = \pi 2.1^2$. Dividing both sides of this equation by $\pi$ yields $b = 4.41$. Therefore, the value of $b$ is $4.41$.

▼ 1.44 — 27

The correct answer is $27$. It's given that a triangular prism has a volume of $216 \text{ cubic centimeters } \text{cm}^3$ and the volume of a triangular prism is equal to $Bh$, where $B$ is the area of the base and $h$ is the height of the prism. Therefore, $216 = Bh$. It's also given that the triangular prism has a height of $8 \text{ cm}$. Therefore, $h = 8$. Substituting $8$ for $h$ in the equation $216 = Bh$ yields $216 = B8$. Dividing both sides of this equation by $8$ yields $27 = B$. Therefore, the area, $\text{in } \text{cm}^2$, of the base of the prism is $27$.

▼ 1.45 — B

Choice B is correct. The area, $A$, of a triangle is given by $A = \frac{1}{2}bh$, where $b$ is the length of a base of the triangle and $h$ is the corresponding height of the triangle. It's given that a triangle has a base length of $10$ centimeters and a corresponding height of $70$ centimeters. Substituting $10$ for $b$ and $70$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}1070$, or $A = 350$. Therefore, the area, in square centimeters, of the triangle is $350$.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.

▼ 1.46 — 126

The correct answer is $126$. It's given that the base of the right rectangular pyramid has a length of $27$ centimeters and a width of $2$ centimeters and that the height of the pyramid is $7$ centimeters. The volume, $V$, of a right rectangular pyramid is given by the formula $V = \frac{1}{3}lwh$, where $l$ and $w$ are the length and width, respectively, of the pyramid's base and $h$ is the height of the pyramid. Substituting $27$ for $l$, $2$ for $w$, and $7$ for $h$ in this formula yields $V = \frac{1}{3}(27)(2)(7)$, or $V = 126$. Therefore, the volume, in cubic centimeters, of the pyramid is $126$.

▼ 1.47 — D

Choice D is correct. Using the volume formula $V = \frac{7\pi k^3}{48}$ and the given information that the volume of the glass is 473 cubic centimeters, the value of k can be found as follows:

$$473 = \frac{7\pi k^3}{48}$$

$$k^3 = \frac{473(48)}{7\pi}$$

$$k = \sqrt[3]{\frac{473(48)}{7\pi}} \approx 10.10690$$

Therefore, the value of k is approximately 10.11 centimeters.

Choices A, B, and C are incorrect. Substituting the values of k from these choices in the formula results in volumes of approximately 7 cubic centimeters, 207 cubic centimeters, and 217 cubic centimeters, respectively, all of which contradict the given information that the volume of the glass is 473 cubic centimeters.

▼ 1.48 — B

Choice B is correct. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(7, 8)$ is halfway between the points $(7, 5)$ and $(7, 11)$ and is a distance of $3$ units from each of those two points. The point $(7, 8)$ is also a distance of $3$ units from $(4, 8)$. Because the point $(7, 8)$ is the same distance from all three given points, it must be the center of the circle. The radius of a circle is the distance from the center to any point on the circle. Since that distance is $3$, it

follows that the radius of the circle is $3$. The circumference of a circle with radius $r$ is equal to $2\pi r$. It follows that the circumference of the circle is $2(\pi)(3)$, or $6\pi$. It's given that the circumference of the circle is $k\pi$. Therefore, the value of $k$ is $6$.

Choice A is incorrect. This is the radius of the circle, not the value of $k$ in the expression $k\pi$.

Choice C is incorrect. This is the $x$-coordinate of the center of the circle, not the value of $k$ in the expression $k\pi$.

Choice D is incorrect. This is the value of $k$ for which $k\pi$ represents the area of the circle, in square units, not the circumference of the circle, in units.

▼ 1.49 — D

Choice D is correct. The area, $A$, of a circle is given by the formula $A = \pi r^2$, where $r$ represents the radius of the circle. It's given that circle $K$ has a radius of $4$ millimeters mm. Substituting $4$ for $r$ in the formula $A = \pi r^2$ yields $A = \pi 4^2$, or $A = 16\pi$. Therefore, the area of circle $K$ is $16\pi$ $\text{mm}^2$. It's given that circle $L$ has an area of $100\pi$ $\text{mm}^2$. Therefore, the total area, $\text{in mm}^2$, of circles $K$ and $L$ is $16\pi + 100\pi$, or $116\pi$.

Choice A is incorrect. This is the sum of the radii, $\text{in mm}$, of circles $K$ and $L$ multiplied by $\pi$, not the total area, $\text{in mm}^2$, of the circles.

Choice B is incorrect. This is the sum of the diameters, $\text{in mm}$, of circles $K$ and $L$ multiplied by $\pi$, not the total area, $\text{in mm}^2$, of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.50 — B

Choice B is correct. If the radius of container A is 16 centimeters and the radius of container B is 25% longer than the radius of container A, then the radius of container B is $16 + (0.25)(16) = 20$ centimeters. The volume of a cylinder is $\pi r^2 h$, where r is the radius of the cylinder and h is its height. Substituting $r = 20$ and $h = 50$ into $\pi r^2 h$ yields that the volume of cylinder B is $\pi(20)^2(50) = 20{,}000\,\pi$ cubic centimeters.

Choice A is incorrect and may result from multiplying the radius of cylinder B by the radius of cylinder A rather than squaring the radius of cylinder B. Choice C is incorrect and may result from multiplying the radius of cylinder B by 25 rather than squaring it. Choice D is incorrect

and may result from taking the radius of cylinder B to be 25 centimeters rather than 20 centimeters.

▼ 1.51 — A

Choice A is correct. The volume, $V$, of a right circular cylinder is given by the formula $V = \pi r^2 h$, where $\pi r^2$ is the area of one of the bases of the cylinder and $h$ is the height. It's given that the area of one of the bases of the cylinder is $81$ square centimeters and that the volume of the cylinder is $2{,}187$ cubic centimeters. Substituting $2{,}187$ for $V$ and $81$ for $\pi r^2$ in the formula $V = \pi r^2 h$ yields $2{,}187 = 81h$. Dividing both sides of this equation by $81$ yields $27 = h$. Therefore, the height, in centimeters, of the cylinder is $27$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.52 — A

Choice A is correct. It's given that circle N has a radius of $7$ mm. The area of a circle is given by the expression $\pi r^2$, where $r$ is the radius of the circle. Substituting $7$ for $r$ in the expression yields $\pi(7)^2$, or $49\pi$, for the area, in $\text{mm}^2$, of circle N. It's also given that the area of circle M is $64\pi$ $\text{mm}^2$. Adding the two areas yields $49\pi + 64\pi$, or $113\pi$, $\text{mm}^2$. Therefore, the total area, in $\text{mm}^2$, of circles N and M is $113\pi$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.53 — C

Choice C is correct. The area, $A$, of a triangle is given by the formula $A = \frac{1}{2}bh$, where $b$ is the base length and $h$ is the height of the triangle. It's given that a triangle has a base length of $56$ centimeters and a height of $112$ centimeters. Substituting $56$ for $b$ and $112$ for $h$ in the formula $A = \frac{1}{2}bh$ yields $A = (\frac{1}{2})(56)(112)$, or $A = 3{,}136$. Therefore, the area, in square centimeters, of the triangle is $3{,}136$.

Choice A is incorrect. This is the value of $56 + 112$, not $(\frac{1}{2})(56)(112)$.

Choice B is incorrect. This is the value of $(\frac{1}{4})(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Choice D is incorrect. This is the value of $(56)(112)$, not $(\frac{1}{2})(56)(112)$.

▼ 1.54 — 31

The correct answer is $31$. The circumference of a circle is equal to $2\pi r$ centimeters, where $r$ represents the radius, in centimeters, of the circle, and the diameter of the circle is equal to $2r$ centimeters. It's given that a circle has a circumference of $31\pi$ centimeters. Therefore, $31\pi = 2\pi r$. Dividing both sides of this equation by $\pi$ yields $31 = 2r$. Since the diameter of the circle is equal to $2r$ centimeters, it follows that the diameter, in centimeters, of the circle is $31$.

▼ 1.55 — C

Choice C is correct. The total volume of the cylindrical can is found by multiplying the area of the base of the can, $75\ \text{cm}^2$, by the height of the can, 10 cm, which yields $750\ \text{cm}^3$. If the syrup needed to fill the can has a volume of $110\ \text{cm}^3$, then the remaining volume for the pieces of
fruit is $750 - 110 = 640\ \text{cm}^3$.

Choice A is incorrect because if the fruit had a volume of $7.5\ \text{cm}^3$, there would be $750 - 7.5 = 742.5\ \text{cm}^3$ of syrup needed to fill the can to the top. Choice B is incorrect because if the fruit had a volume of $185\ \text{cm}^3$, there would be $750 - 185 = 565\ \text{cm}^3$ of syrup needed to fill the can to the top. Choice D is incorrect because it is the total volume of the can, not just of the pieces of fruit.